QPCODE: **Reg.No : .....**

Name : .....

MAHATMA GANDHI UNIVERSITY, KOTTAYAM
MGU-UGP (HONOURS) EXAMINATION
IV SEMESTER
MG4SECCFS200: PROGRAMMING IN JAVA
SEC
(2024 ADMISSION ONWARDS)

Duration: 1.5 Hrs. **Maximum Marks: 50 Marks**

Remember(K), Understand(U), Apply(A), Analyse(An), Evaluate(E), Create(C), Skill(S), Interest(I) and Appreciation(Ap)

Students should attempt at least one question from each course outcome to enhance their overall outcome attainability.

Part A

Short Answer Type Questions

Answer any **10** questions

Each question carries **1** mark

1. Explain why Java is called a platform-independent language [CO:1]

2. Distinguish between the roles of the JDK and the JRE. [CO:1]

3. Explain the difference between `/` and `%` operators in Java? [CO:1]

4. Which data type is best suited to store a single Unicode character? [CO:1]

5. Describe the primary difference between a `while` loop and a `do-while` loop in Java. [CO:1]

6.	How does the `continue` statement differ from the `break` statement when used inside a `for` loop?	[CO:1]
7.	Explain purpose of a `try-catch` block in Java exception handling?	[CO:2]
8.	Describe the finally keyword in java	[CO:2]
9.	What does it mean when a thread is in the "Runnable" state?	[CO:2]
10.	What is the default layout of an Applet?	[CO:1]
11.	**State** the difference between `drawOval()` and `fillOval()` ?	[CO:2]
12.	Which method is used to update the text displayed by a `JLabel` after it has been created?	[CO:3]

[1x10 = 10]

Part B

Short Essay Type Questions

Answer any **4** questions

Each question carries **5** marks

13.	Explain constructor overloading in Java.	[CO:1]
14.	Differentiate between `static` methods and instance methods.	[CO:1]
15.	Explain the mechanism of catching multiple exceptions with an example.	[CO:2]
16.	Discuss the Methods Affecting Thread State Transitions.	[CO:2]
17.	Examine the event-handling mechanism of a JButton	[CO:3]
18.	Analyze the Advantages of Swing Components over AWT Components	[CO:3]

[5x4 = 20]

Part C

Essay Type Questions

Answer any **2** questions

Each question carries **10** marks

19.	Discuss the behavior of access modifiers (`private` , `protected` , `public`) in Java	[CO:1]
20.	What are the two primary ways to create a thread in Java? Explain with an example	[CO:2]
21.	Explain the Delegation Event Model in Java in detail.	[CO:3]

[10x2 = 20]